Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

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Abstract

Classical molecular dynamics is a powerful method of theoretically describing the properties of a material but is often inconsistent with quantum molecular dynamics. Quantum molecular dynamics, while more accurate, are more computationally intensive and not feasible for systems of more than 1000 atoms. Machine learning is used to bridge the gap between quantum accuracy and the scalability of classical interatomic potentials. We report on progress made testing the performance of a quantum-accurate classical interatomic potential for gold against state-of-the-art classical potentials in describing experimentally verified properties of gold using LAMMPS (Large Scale Atomic/Molecular Massively Parallel Simulator). The elastic constants predicted by the machine learning model were a Young's modulus of 74.1 GPa, bulk modulus of 105.2 Gpa, shear modulus of 26.8 GPa, and a Poisson's ratio of 0.38. The elastic constants predicted by the industry standard model were a Young's modulus of 35.7 GPa, bulk modulus of 166.8 Gpa, shear modulus of 12.2 GPa, and a Poisson's ratio of 0.46. Future work includes extending this study to simulate rare events in larger systems, such as the herringbone reconstruction of the Au(111) surface, that are not accurately described by current classical potentials.

Introduction

Classical molecular dynamics is a numerical method in which you solve the equations of motion for a small representative system in order to gain insights into macroscopic properties of a material.

Molecular dynamics is widely used in scientific fields as well as industry due to its cost efficiency, and versatility. Geologists use molecular dynamics to study the materials in the earth's mantle that we do not have means to measure physically[1]. Alloys for planes are planned and tested using molecular dynamics[13]. Theoretical materials are designed and tested as deep-ultraviolet light sources using molecular dynamics.[9]

Machine learning and deep learning is being used to model quantum accurate effects in metals[6]. The computational complexity of various models has been compared and benchmarked[17]. Machine learning has been used to model the herringbone reconstruction of gold, using DeepMD, a Tensorflow based machine learning package [8]. This is an experimentally observed phenomena that has properties of interest such as surface chemistry, catalysis, and nanostructure growth.

Molecular Dynamics as a numerical method makes two major assumptions, that we can approximate complicated atoms as particles, and that these particles can then predict the macroscopic properties of a thermodynamic system. The approximation of atoms as particles is supported by the thermal De Broglie wavelength, Born-Oppenheimer approximation, and Ehrenfest's theorem. The Thermal De Broglie Wavelength is a measure of uncertainty in the position of a particle, given by,

$$\lambda_{th} = \sqrt{\frac{2\pi\hbar^2}{mk_BT}}. \tag{1}$$

At low temperatures or velocities and in particles with little mass such as hydrogen, the thermal De Broglie wavelength becomes significant. However in large quantum numbers and reasonable temperatures, this uncertainty is dominated by the classical thermal fluctuations.

The Born-Oppenheimer approximation states that the electrons of an atom are much less massive, and therefore move much faster than the more massive atomic core. Because of this we can effectively separate the dynamics of the electrons from those of the atomic core. This is a core idea to density function theory (DFT), and allows us to build interatomic potentials using electron forces.

Ehrenfest's Theorem states that quantum expectation values mimic classical dynamics in large quantum numbers and systems,

$$m\frac{d}{dt}\langle\vec{r}\rangle = \langle\vec{p}\rangle. \tag{2}$$

The expectation value of the position operator $\vec{r}$ is related to the expectation value of momentum, $\vec{p}$. This can be rearranged to give,

$$m\frac{d^2\langle\vec{r}\rangle}{dt^2} = -\left\langle\nabla U(\vec{r})\right\rangle. \tag{3}$$

At large quantum numbers, this will closely approximate the classical statement of Newton's second law,

$$m\frac{d^2}{dt^2}\langle\vec{r}\rangle \approx -\nabla U(\langle\vec{r}\rangle). \tag{4}$$

This allows us to approximate closely the quantum mechanical effects of atoms using classical potentials. To support the claim that we can approximate macroscopic properties using smaller representative particle systems, we introduce the statistical ensembles. statistical ensembles are collections of microstates, following macroscopic properties or observables, such as temperature, number of particles, volume, or pressure.

The Ergodic hypothesis states that the time average of one particle as it approaches infinite time will be equal to the ensemble average of a system,

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T A(t)\,dt = \langle A\rangle_{ensemble}, \tag{5}$$

if a system is ergodic. Because it is not practical to observe a particle for infinite time, we approximate the ensemble average using a collection of particles over a limited amount of time. In a reasonably ergodic system, this value is well converged, allowing us to use small systems and limited times. [2]

Molecular Dynamics in this experiment will be done in the microcanonical ensemble, the canonical ensemble, and the isobaric ensemble. In the microcanonical ensemble (NVE), the number of particles, volume, and energy of a system are held constant. The equations of motions in this ensemble are solved using Newton's second law,

$$\sum \vec{F} = m\vec{a}. \tag{6}$$

The forces are found as the negative gradient of the potential energy of a particle,

$$\vec{F_i} = -\nabla_i U(\vec{r_1}, \ldots, \vec{r_n}), \tag{7}$$

that is dependent on the position vectors of each particle in the system.

In the canonical ensemble (NVT), number of particles, volume, and temperature are held fixed. In the isobaric (NPT), number of particles, pressure, and temperature are held constant. In both ensembles, the equations of motions are solved using the Langevin equation,

$$m\frac{d^2\vec{r_i}}{dt^2} = -\lambda\frac{d\vec{r_i}}{dt} + \eta(t) - \nabla_i U(\vec{r_1}, \ldots, \vec{r_n}). \tag{8}$$

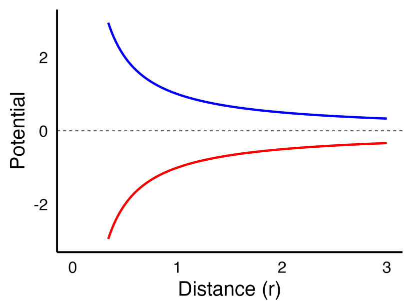
The Langevin equation introduces two additional terms to account for the temperature or pressure, a friction-like term based on the first derivative of position, and a 'kick' term that gives the system a velocity distribution based on the Boltzmann distribution to simulate temperature. These terms are used as a 'thermostat' for computational experiments, to keep constant pressure or temperature, the velocities of the particles are changed based on this distribution.

Molecular dynamics simulation methods differ in the handling of this potential energy term, called a potential. The potential is generally a function dependent on the chemical identity of the particle in question, and the other particles in the system. There are general forms of potentials[11], but in most cases the potential developed is unique to the material simulated.

A simple example of the potential term is the pair potential. The potential energy term can be separated into the contributions of the single body forces, such as gravity or external magnetic field, and pair interactions, giving,

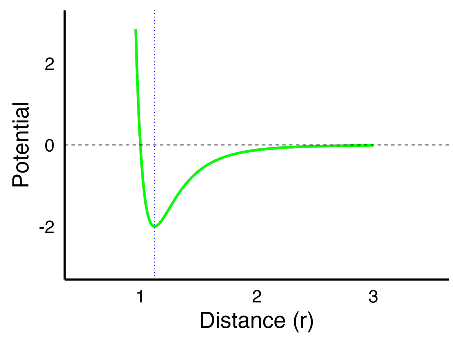
$$U_{tot} = \sum_i U_i + \frac{1}{2} \sum_{i,j} U(r_{i,j}). \tag{9}$$

Pair potentials focus on this pair interaction, the two most common examples of this are Coulomb potentials and Lennard-Jones potentials shown in Figure 1.



$U_C(r_{i,j}) = \frac{q_i q_j}{4\pi\epsilon_0 r_{i,j}}$

Equal charges repel (top line) and opposite charges attract (bottom line). Forces are stronger as distance is shorter and tend towards 0 as distance gets large.



$U_{LJ}(r_{i,j}) = 4\epsilon \left[\left(\frac{\sigma}{r_{i,j}} \right)^{12} - \left(\frac{\sigma}{r_{i,j}} \right)^{6} \right]$

At very short distances the potential strongly repels, far away it trends towards no interaction. The dotted line indicates the equilibrium distance where the force derived from the potential is 0.

Figure 1: Coulomb potential (left) and Lennard-Jones potential (right)

The coulomb potential gives a repulsing force when charges are equal, and an attractive force when charges are opposite. The Lennard-Jones potential gives a repulsive force at short distances, and an attracting one further away, while similarly to the Coulomb potential diminishing at large distances. The Lennard-Jones potential has an equilibrium distance where the force will be zero.

The Embedded Atom Method introduces a potential function that describes the energy required to place a reference electron into a host electron density,

$$E_{tot} = \sum_i f(\bar{\rho}_i) + \frac{1}{2} \sum_{i,j \quad j \neq 1} U_{i,j}(r_{i,j}) + E_{error}. \tag{10}$$

The electron density in this term is spherically averaged,

$$\bar{\rho}_i = \sum_j^N \rho(r_{i,j}). \tag{11}$$

This function is semi-empirically determined[3]. Values of various physical properties can be calculated through physical experiments or first principles calculations and tabulated. The function is built from these values, commonly through the use of cubic spline interpolation. The potential used in this experiment is based on first principles calculations of the elastic tensor of gold.

Embedded Atom Method potentials closely model many physical properties, without a significant increase in computation time over pair potentials[3], making it the industry standard method for metals such as gold.

The model does not properly describe quantum mechanical, or directional effects, due largely to the spherical averaging of the electron density. Machine learning potentials begin from deconstructing the total potential energy of a system into a sum of its individual contributions for a particle, and then it deconstructs those potential energies into contributions from 'basis functions,'

$$E_{tot} = \sum_i U_i = \sum_i \sum_\alpha c_\alpha B_\alpha^{(i)}. \tag{12}$$

The general form of the machine learning potential used in this experiment will be the atomic cluster expansion method[5]. In the generalized atomic cluster expansion method, any invariant descriptor inherent to an atom's structure can be used to build the basis functions. The constant term attached to each basis function is fit using a neural network, trained on first principles data. A common method used, and the one used in this experiment, is to use the polyspectra[4] as the basis functions.

The polyspectra decomposes the electron wave function into 'modes' or multi-body contributions. The first-order term in this expansion is the power spectrum, this experiment's model uses up to the third order bispectrum component. This is a 3-body term that allows the model to encode angular dependence, improving on the disadvantage of embedded atom method of not encoding directional dependence in the potential energy function. In general, additional terms can be added to this model, increasing its accuracy and at the same time its computational complexity. For purposes of this experiment, the third-order term is sufficient to model surface effects.

The goal of the machine learning models is to be quantum accurate, with the scalability of classical potentials. We will determine if they are accurate and scalable to the same use cases as the industry standard, and in future work extend the working model to experimentally observed phenomena not accurately modeled by the industry standard, such as the herringbone reconstruction[8]. To benchmark the models, we will use well known mechanical properties of gold. This allows us to compare the machine learning model with the industry standard embedded atom method model in recreating physical data.

Stress-strain tests are commonly done on room temperature metals[7]. In uniaxial tension tests, the metal is stretched along one of its axis, and pressure and distance stretched are recorded. From this test we can gain insights into the behavior of the metal through stress strain curves as in Figure 2. Young's

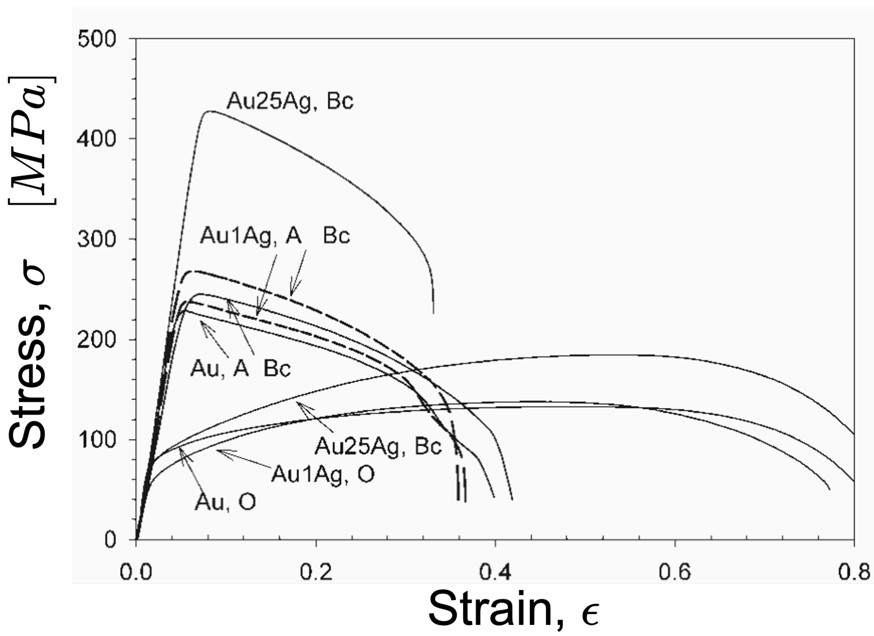


Figure 2: Stress-strain curve of physical gold and alloys. Young's modulus is given by the slope of the linear portion of each line. The elastic limit is the point where linearity in that line breaks.

modulus (E) is calculated at the slope of the linear portion of this curve,

$$.E = \frac{\sigma}{\varepsilon} \tag{13}$$

The elastic limit refers to the point at which the material breaks, relieving stress by deforming in an inelastic way. The elastic limit of gold is 205MPa. Compression tests are used in conjunction with the tensile tests to obtain the bulk modulus (K), shear modulus (G), and Poisson's ratio (ν) of the material. Young's modulus describes the resistance of the material to stretching; The bulk modulus describes its resistance to compression,

$$K = \frac{\Delta P}{-\frac{\Delta V}{V}}. \tag{14}$$

The shear modulus describes resistance to shear or twisting forces,

$$G = \frac{\tau}{\gamma}. \tag{15}$$

Poisson's ratio describes the rate at which the other axis move in or out when the material is stretched or compressed along an axis,

$$\nu = -\frac{\varepsilon_{\mathrm{transverse}}}{\varepsilon_{\mathrm{axial}}}. \tag{16}$$

Gold at room temperature has a face-centered cubic (FCC) lattice and lattice parameter of 4.078Å shown in Figure 3. In future work, the herringbone surface

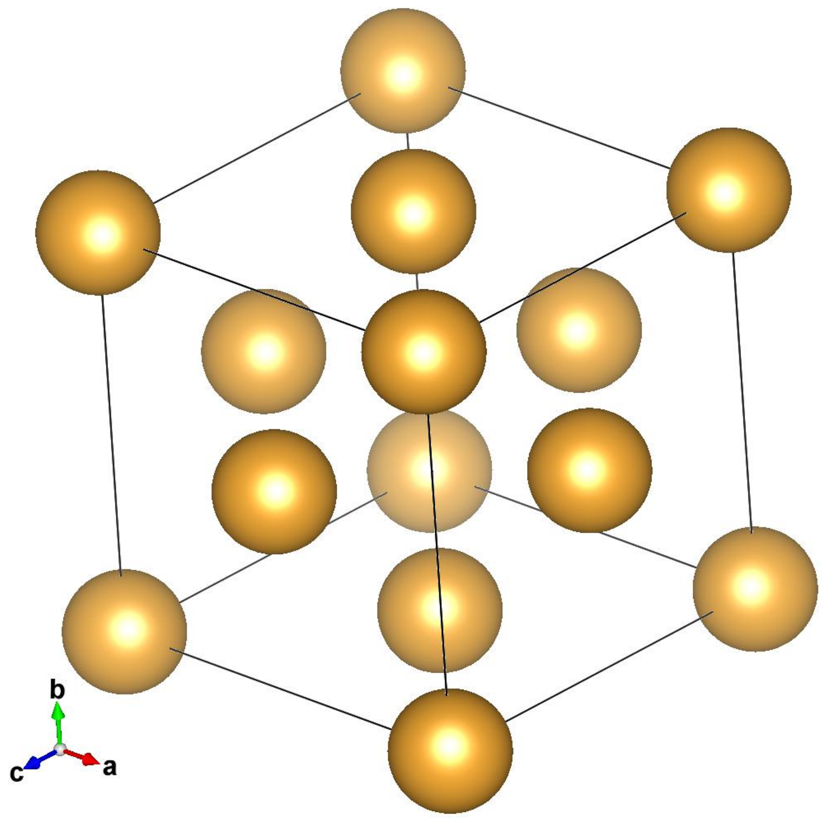


Figure 3: Gold FCC unit cell with lattice parameter 4.078Å

reconstruction occurs when the surface crystal lattice of gold transitions to a hexagonal close packed lattice (HCP) in order to relieve strain, by compressing uniaxially by roughly 4%.

The radial distribution function, shown in Figure 4, is a measure representative of the probability of finding a particle at a given distance. It is used in physical and computational experiments to analyze the ordering of particles in a system. It counts the number of particles a given distance from a reference particle, and averages over all particles and configurations. If particles are in an ordered structure, more particles will be found at certain distances; and if they are disordered, they will be spread out. Sharp peaks correspond to crystalline structures, broad peaks describe amorphous or liquid behavior, and a straight line would be an ideal gas. We test the computation efficiency of machine learn-

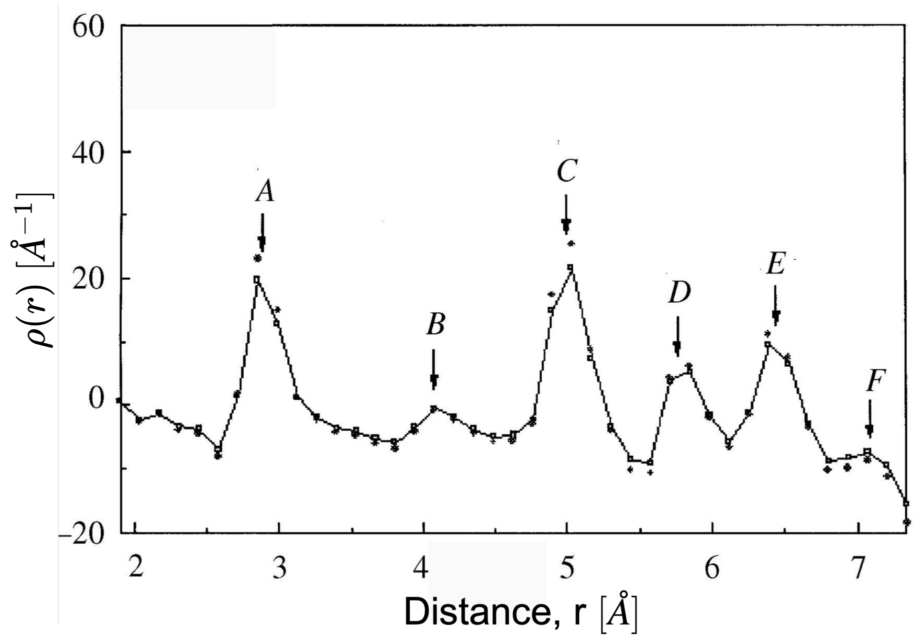


Figure 4: Radial distribution function of physical gold[10]

ing potentials against the industry standard embedded atom method for metals in modeling mechanical properties of gold. We test its accuracy, and scalability of computation.

Methods

LAMMPS[15] (Large Scale Atomic/Molecular Massively Parallel Simulator) was used to perform the classical molecular dynamics simulations. LAMMPS numerically integrates Newton's equations for motion for a collection of simulated particles. Particles are placed in a simulation box with defined positions, velocities, and masses. Parameters including boundary conditions, thermodynamic ensembles, and the potentials are set and the system is allowed to evolve. This refers to numerically integrating the system through set time steps based on the given conditions. We used the FITSNAP[14] molecular machine learning package for LAMMPS. FITSNAP creates a force field potential from training data that can be used in LAMMPS to simulate large systems. Figure 5 shows the process in which FitSNAP takes in training data and forms the potential. These steps were followed using each potential. We begin with a unit cell of

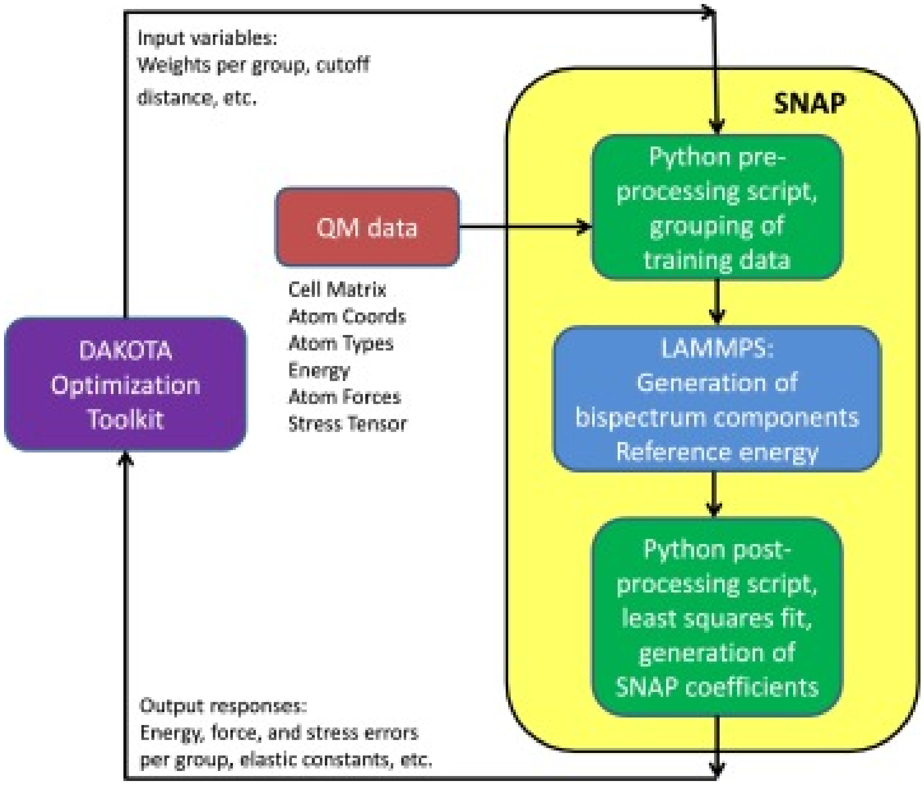


Figure 5: FitSNAP machine learning process

gold at room temperature which is in a face centered cubic (FCC) lattice with lattice parameter 4.078Å shown in Fig. This cell is then duplicated to create a bulk crystal 10 unit cells in each direction. To ensure the simulation explores a phase space trajectory that is mostly ergodic and representative of the larger ensemble, the system must be equilibrated. This allows the system to forget non-physical conditions such as the initial configuration of stationary atoms on a lattice. To quickly approach equilibrium, the potential function is first minimized using the conjugate gradient method. The system is then allowed to evolve under the NPT ensemble at room temperature, this is known as relaxation. Annealing is used when the system exhibits non-ergodic conditions, the system is heated to high temperatures and allowed to cool slowly while evolving similarly to the relaxation process. This can help overcome potential energy barriers that can force non-ergodic conditions on a system. Using LAMMPS the equilibrated structure is stretched along one axis in the isobaric (NPT) ensemble until fracture. The lengths and pressure tensor components of the system are calculated in LAMMPS.

Results

Table 1 shows the calculated constants of physical gold[10] and the calculated values from the EAM and ACE models.

Table 1: Elastic constant of gold from physical measurement, EAM model, and ACE model.

	Actual	EAM	ACE
Young's Modulus	78.5	35.7	74.1
Bulk Modulus	177.6	166.8	105.2
Shear Modulus	26.0	12.2	26.8
Poisson's Ratio	0.44	0.46	0.38

Figure 6 shows the stress-strain curve of bulk gold using the embedded atom method. Its shape qualitatively follows the same behavior as the physical gold tests. It predicts an elastic limit of over 4 GPa, and a young's modulus of 34.5

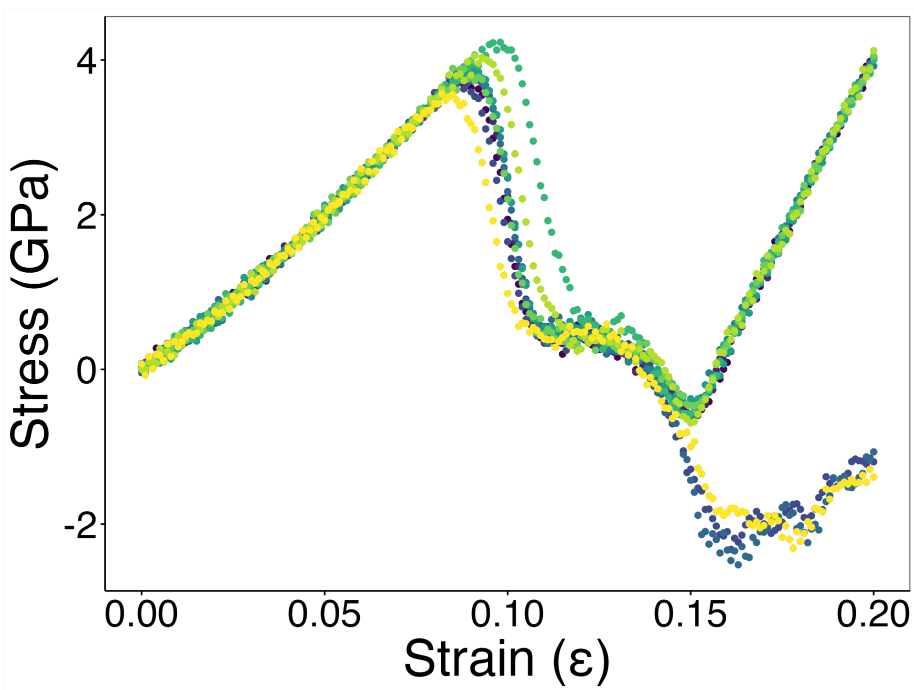


Figure 6: Embedded atom method stress-strain curve

as shown in figure 7. Figure 8 shows the stress-strain curve of the atomic cluster expansion model. It more closely matches the elastic limit of 205 MPa where its linear portion ends at roughly 250 MPa, and has a more accurate young's modulus of 74.5. Figure 8 however shows no relief of stress in the continual in-

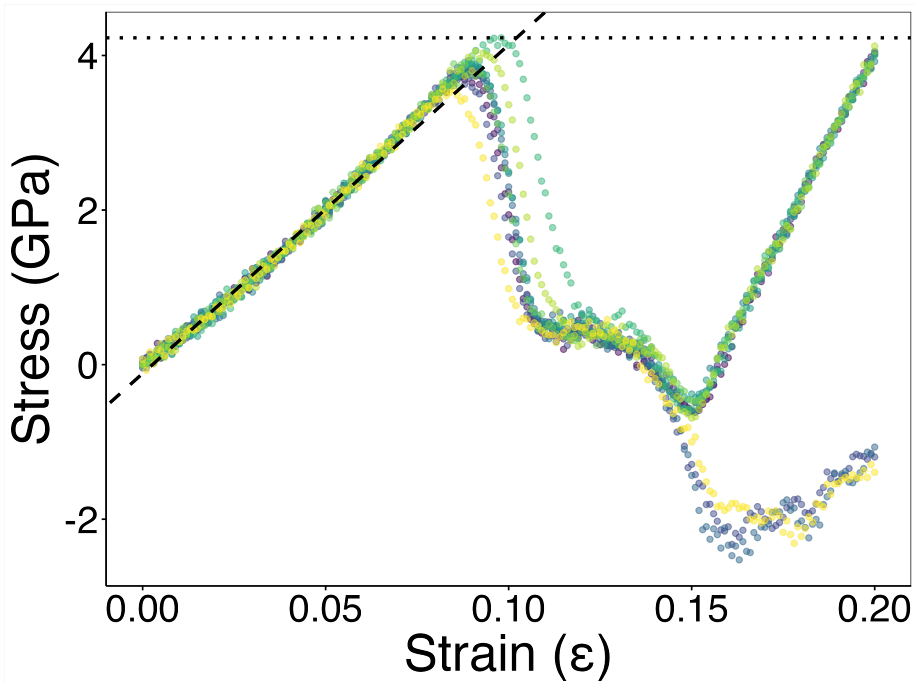


Figure 7: Embedded atom method stress-strain curve showing the elastic limit (dotted line) and the slope giving Young's modulus (dashed line)

crease of strain, implying amorphous behavior. The radial distribution function is used to verify the behaviors of each model, shown in Figure 10. Each model shows sharp crystalline peaks, until temperatures are increased in the atomic cluster expansion models. Figures 9a and 9b show crystalline peaks as expected, 9c and 9d both show amorphous behavior when temperature is introduced to the atomic cluster expansion model.

Discussion

The embedded atom method model fails to properly describe elastic constants and behavior of gold, this is due to its shortcomings in describing directional and quantum mechanical effects. The atomic cluster expansion model accurately models the elastic behaviors of gold at low strain and temperature. When temperature or increased strain is introduced the model behaves erratically, predicting a liquid gold at room temperature. This happens because the initial training data was done at 0K. The model must be retrained using sampled data of perturbed lattices representative of 10K-3000K gold. The model shows potential however in that the changed training data will only change a constant factor in front of each basis function. Retraining the test data will

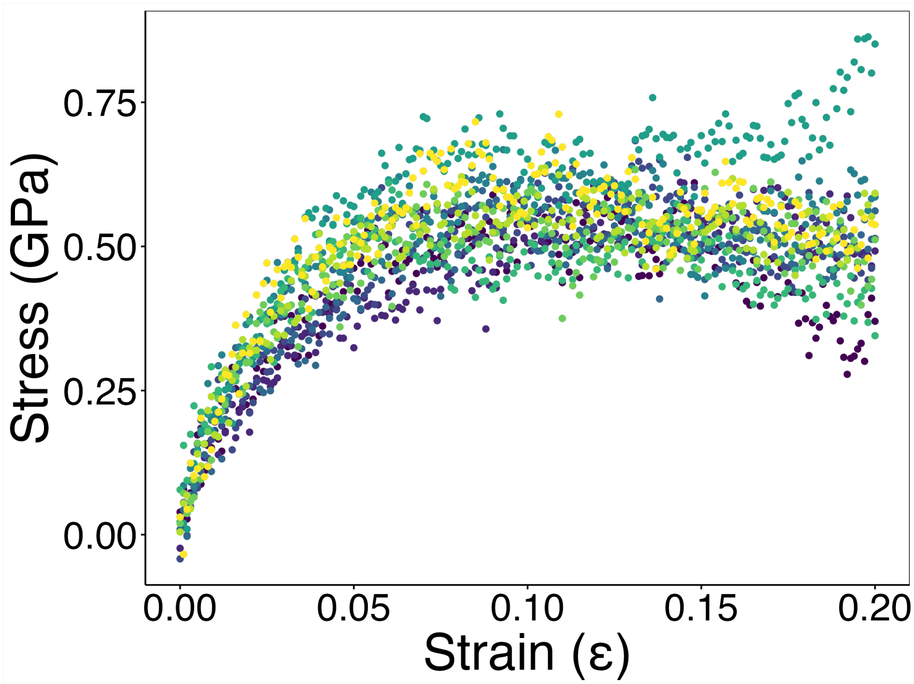


Figure 8: Atomic cluster expansion Stress-strain curve

not raise the computational complexity of the model. It is predicting quantum accurate data, just not the proper data for gold.

Conclusion

We benchmarked the effectiveness of a machine learning interatomic potential for gold against the industry standard embedded atom method, as well as physical data. The elastic constants predicted by the ACE model were a Young's modulus of 74.1 GPa, bulk modulus of 105.2 Gpa, shear modulus of 26.8 GPa, and a Poisson's ratio of 0.38. The elastic constants predicted by the EAM model were a Young's modulus of 35.7 GPa, bulk modulus of 166.8 Gpa, shear modulus of 12.2 GPa, and a Poisson's ratio of 0.46. The model improves the predictions of elastic constants at low temperature but fails when significant temperature or strain are introduced, this is due to the lack of training data using perturbed lattices. Our next steps will be to retrain the neural network using perturbed lattices corresponding to a distribution in 10K-3000K gold. Upon satisfactory completion of mechanical property simulations we will extend the model to less well known phenomena such as the herringbone surface reconstruction. In future experiments we also plan to benchmark the Spectral Neighbor Analysis Potential (SNAP) against the Atomic Cluster Expansion (ACE) potential for

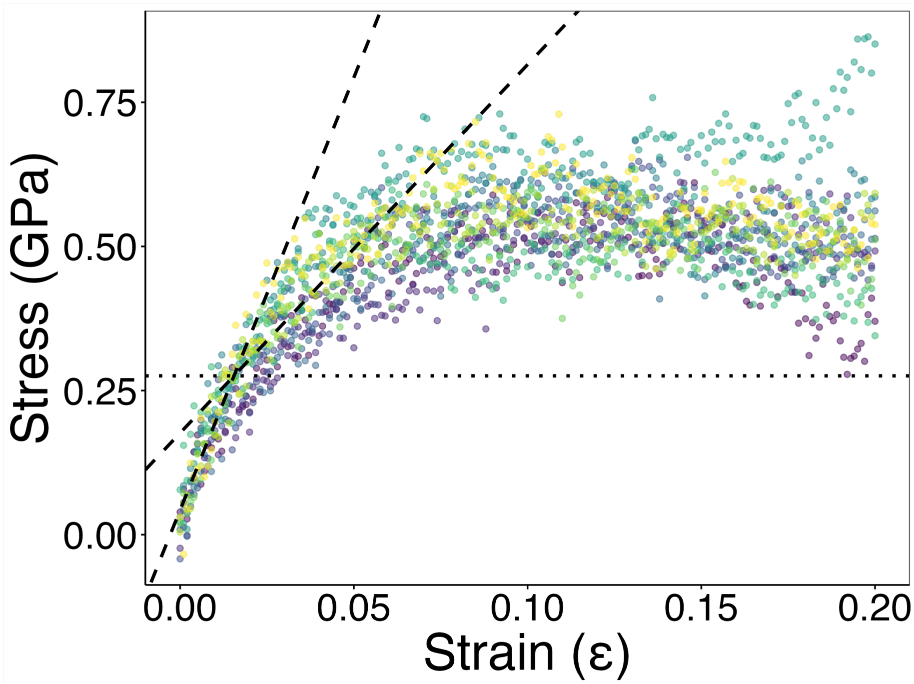
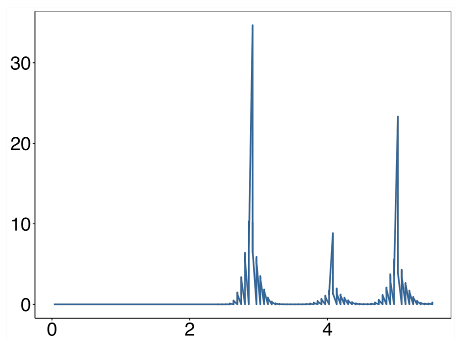


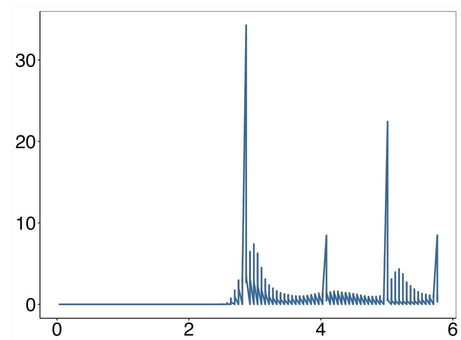
Figure 9: Atomic cluster expansion Stress-strain curve Showing Elastic Limit (dotted line) and best fit lines (dashed) of 0-0.015 strain, and 0.015-0.05 strain

gold.

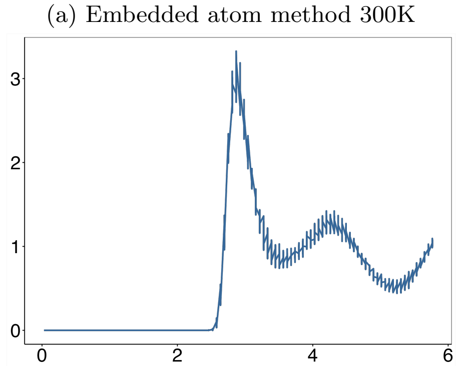
To simulate the herringbone reconstruction, we begin by building a stripe reconstruction unit cell from density functional theory calculations[16]. This is then duplicated along each axis until the periodic length of 30 nm[8] of the herringbone reconstruction is reached, and then grains of this reconstruction are rotated to mimic the observed effect using Atomsk[12]. LAMMPS is then used to calculate the energies of the surfaces of this structure under varying conditions, by showing the structure is energetically favorable over its bulk form under a certain condition, we gain insight into what is causing the phase change to happen. If the effect is unstable or not energetically favorable under reasonable conditions, then the model may need reworked to include higher order terms through basis functions.



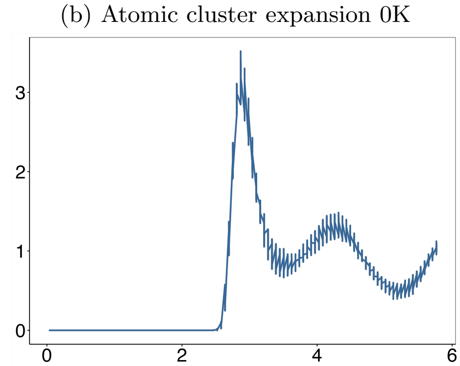
(a) Embedded atom method 300K



(b) Atomic cluster expansion 0K



(c) Atomic cluster expansion 300K after relaxation



(d) Atomic cluster expansion 300K after annealing

Figure 10: Radial Distribution Functions

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